TÜRKİYE CUMHURİYETİ İSTANBUL TEKNİK ÜNİVERSİTESİ

REPUBLIC OF TURKEY ISTANBUL TECHNICAL UNIVERSITY

Muhammed Kocabaş

BİLGİSAYAR VE BİLİŞİM FAKÜLTESİ
BİLGİSAYAR MÜHENDİSLİĞİ Lisans Programının tüm gerekliliklerini HAVA HARP OKULU'nda başarı ile tamamlayarak 669 sayılı KHK kapsamında

Bilgisayar Mühendisi

unvanını bütün onur, yetki ve sorumluluklarıyla birlikte almaya hak kazanmıştır.

having satisfactorily fulfilled all the requirements of the Undergraduate Program, as in COMPUTER ENGINEERING Program of the FACULTY OF COMPUTER AND INFORMATICS, at the AIR FORCE MILITARY ACADEMY in accordance with the decree law numbered 669

has been awarded the degree of

Bachelor of Science

in COMPUTER ENGINEERING

with all the rights, privileges and honors thereto appertaining.

Prof. Dr. Uğur BAYAZIT
Dekan / Dean

İSTANBUL TEKNİK ÜNİVERSİTESİ 1773

Prof. Dr. Mehmet KARACA
Rektör / Rector

Diploma No: 2016-09-0239
Mezuniyet Tarihi/Date of Graduation:29.07.2016

T.C.Kimlik Numarası / Turkish Identification Number : 12402061602

Yıldız BÜYÜKÇOLAK
Öğrenci İşleri Daire Başkanı / Registrar

DIPLOMA SUPPLEMENT

Istanbul Technical University
Diploma Supplement

1. INFORMATION IDENTIFYING THE HOLDER OF THE QUALIFICATION

1.1. Family name(s) : Kocabaş
1.2. Given name(s) : Muhammed
1.3. Date of birth (day/month/year) : 16.05.1994
1.4. Student identification number or code : 1310

2. INFORMATION IDENTIFYING THE QUALIFICATION

2.1. Name of the qualification : Bilgisayar Mühendisliği
Title conferred : Bilgisayar Mühendisi

2.2. Main field(s) of study for the qualification:
Computer Engineering

2.3. Name and status of awarding institution
İstanbul Teknik Üniversitesi, Devlet Üniversitesi
Bu belge 669 sayılı KHK gereğince düzenlenmiştir.
Istanbul Technical University, State University
This document has been issued in accordance with the decree law numbered 669.

2.4. Name and status of institution administering studies
Same as 2.3

2.5. Language(s) of instruction / examination
Turkish

3. INFORMATION ON THE LEVEL OF THE QUALIFICATION

3.1. Level of qualification:
First Cycle(Bachelor's Degree)

3.2. Official length of programme:
4 years, 2 semesters per year, 17 weeks per semester, 240 ECTS

3.3. Access requirement(s):
Diploma and through medical check-up for the Military High School graduates. High School Diploma from Civilian High Schools, adequate score on Centralized National University Placement Examination, oral exam, through medical check-up, sports test and student flight training selection.

4. INFORMATION ON THE CONTENTS AND RESULTS GAINED

4.1. Mode of study:
Full - time

4.2. Programme requirements:
The Bachelor's Degree is awarded to students who have successfully completed the compulsory and elective courses of the curriculum and have obtained a minimum cumulative grade average (CGPA) of 2.00/4.00.

Objectives
The goal of the program is to equip students with the advanced level of knowledge and understanding of concepts, theories and techniques used in their fields.

Key program outcomes
Being able to think critically about problems and events using analyze analysis techniques with the help of theoretical knowledge, to come up with solutions, to have the ability to prepare outline of the programs and decisions.

Being able to have the awareness of lifelong learning; within this scope, continuously being able to follow up-to-date developments in the area in order to refresh herself/himself.
Being able to study individually and in well-disciplined teams, to have the responsibility of using leadership abilities.
Being able to run a project, to be aware of the knowledge of workplace applications, health of personnel, environment and work safety.
Being able to communicate efficiently both in written and oral Turkish.
Being able to speak at least one foreign language at general European Language Portfolio B1 level.

Faculty of Computer and Informatics Engineering

DIPLOMA NO AND DATE

2016-09-0239
29 July 2016

Following the model developed by:

*UNESCO - CEPES
*COUNCIL OF EUROPE
*EUROPEAN COMMISSION

The Diploma Supplement follows the model developed by European Commission, Council of Europe and UNESCO/CEPES.

The purpose of the supplement is to provide sufficent independent date to improve the International "transparency" and fair academic and professional recognition of qualifications (diplomas, degrees, certificates, etc.). It designed to provide a description of the nature, level, context, content and status of the studies that were pursued and successfully completed by the individual named on the original qualification to which this supplement is appended. It should be free from any value-judgments, equivalence statements or suggestions about recognition

Information in all eight sections should be provided. Where information is not provided, an explanation should give the reason why.

4.3 Program Details:(e.g.modules or units studied), and the individual grades / marks / credits obtained:

SEMESTER I

CODE	COURSE TITLE	COURSE CATEGORY	LOCAL CREDIT	ECTS CREDIT	GRADE
BLG 100	INTRO.TO COMPUTER	REQUIRED	0,5	2,0	AA
HRT 100	MAPPING	REQUIRED	2,0	2,0	AA
MAT 101	MATHEMATICS I	REQUIRED	4,0	8,0	AA
İNG 101-I	ENGLISH I (I)	REQUIRED	12,0	18,0	BA

SEMESTER II

CODE	COURSE TITLE	COURSE CATEGORY	LOCAL CREDIT	ECTS CREDIT	GRADE
HVG 100	AVIATION AND WAR HISTORY	REQUIRED	2,5	2,0	AA
BLG 101	COMPUTER PROG.	REQUIRED	3,5	6,0	AA
MAT 102	MATHEMATICS II	REQUIRED	4,0	7,0	CB
FZK 101	PHYSICS I	REQUIRED	4,0	6,0	AA
İNG 102-S	ENGLISH II (S)	REQUIRED	7,0	9,0	AA

SEMESTER III

YÖN 100	BEHAVIOR SCIENCES	REQUIRED	2,0	2,0	AA
ELK 200	FUND OF ELEC ENG	REQUIRED	2,0	3,5	CB
HVC 285	COMP AIDED TECH.DRAW.	REQUIRED	2,5	4,0	AA
İNG 201-S	ENGLISH S	REQUIRED	2,5	2,5	AA
TRK 200	TURKISH	REQUIRED	3,0	3,0	AA
BLG 211	DATA STR AND ALGORTHM	REQUIRED	3,5	5,5	AA
MAT 201	LINEAR ALGEBRA	REQUIRED	4,0	5,0	CC
FZK 102	PHYSICS II	REQUIRED	4,0	6,0	AA

SEMESTER IV

YÖN 200	ECONOMICS	REQUIRED	2,0	2,0	AA
SOS 100	SOCIOLOGY	REQUIRED	2,0	2,0	AA
KİM 100	CHEMISTRY	REQUIRED	2,5	4,0	AA
İNG 202-S	ENGLISH IV	REQUIRED	2,5	2,5	AA
BLG 381	DISCRETE MATHEMATICS	REQUIRED	3,0	3,5	AA
ELK 210	FUNDAMENTAL ELECTRONICS	REQUIRED	3,0	3,0	AA
BLG 202	OBJECT ORIENTED PROG	REQUIRED	3,0	4,0	AA
END 250	STATISTICS AND PROBABILITY	REQUIRED	3,0	4,0	BA
MAT 202	DIFFERANTIAL EQUATIONS	REQUIRED	4,0	5,0	DC

SEMESTER V

ELK 316	ELECTRONICS LAB.	REQUIRED	1,5	2,0	AA
İTA 301	ATA PRIN.HIST.TUR REV.I	REQUIRED	2,0	2,0	AA
İNG 3819.2	ENGLISH 3819.2	REQUIRED	2,0	2,0	AA
HVC 282	MATERIAL SCIENCE	REQUIRED	2,0	4,0	AA
BLG 452	ALGORTHM ANALYSIS	REQUIRED	3,0	4,0	AA
ELK 315	DIGITAL ELECTRONICS	REQUIRED	3,0	4,0	AA
BLG 473	INF.WAR. & CYB.SEC.	REQUIRED	3,0	4,0	AA
BLG 446	INTERNET PROGRAMMING	REQUIRED	3,0	4,0	AA
END 305	OPERATIONAL RESEARCH	REQUIRED	3,0	4,0	BB

SEMESTER VI

İNG 3820.2	ENGLİSH 3820.2	REQUIRED	1,5	2,0	AA
İTA 302	ATA PRIN.HIST.TUR REV.II	REQUIRED	2,0	2,0	BA
HRK 300	MILIT.OPRS.MNGMNT.-II	REQUIRED	2,5	2,5	BA
BLG 431	COMPILER DESIGN	REQUIRED	3,0	4,0	AA
HVC 389	DIGITAL ANALYSIS	REQUIRED	3,0	4,0	AA
BLG 372	COMPUTER ORGANIZATION	REQUIRED	4,0	5,0	BB
BLG 321	INTO.TO DATABASE SYS	REQUIRED	4,0	5,0	AA
BLG 332	OPERATING SYSTEMS	REQUIRED	4,0	5,5	AA

SEMESTER VII

BLG 472	MIC PROCES&MIC COMP LAB	REQUIRED	1,0	2,0	BA
İNG 4921.2	ENGLİSH 4921.2	REQUIRED	1,5	2,0	AA
SYT 400	POLITICAL HISTORY	REQUIRED	2,0	2,0	AA
TRK 400	TURKISH II	REQUIRED	2,0	3,0	AA
HVC 381	INTR.TO AVIA SCI-I	REQUIRED	3,0	4,0	AA
BLG 471	MIC PROCES&MIC COMP	SELECTIVE	3,0	4,0	AA
BLG 451	SOFTWARE ENGINEERING	REQUIRED	3,0	4,0	AA
SOS 400	STRATEGİC AND İNTER. RELATİONS	REQUIRED	3,0	3,0	AA
BLG 441	COMPUTER NETWORK	REQUIRED	4,0	6,0	AA

SEMESTER VIII

İNG.4922.2	ENGLISH 4922.2	REQUIRED	2,0	2,0	AA
HUK 400	LAW	REQUIRED	2,0	2,0	BB
YÖN 300	MAN AND ORG	REQUIRED	2,0	3,0	AA
BLG 463	ARTIFICIAL INTELLIGENCE	SELECTIVE	3,0	4,0	AA
BLG 444	COMPUTER ENCODING	REQUIRED	3,0	4,0	AA
BLG 494	GRADUATION PROJECT	REQUIRED	3,0	7,0	AA
HVC 482	INTR.TO AVIA SCI II	REQUIRED	3,0	4,0	AA
BLG 353	SYSTEM ANALYSIS	REQUIRED	3,0	4,0	AA

Total Credits : 178.0 240

Cumulative Grade Point Average : 3.74

4.4. Grading scheme and grades

The official grading system uses letter grades. Passing grades range from AA to DD; F is failing. Cumulative Grade Point Average (CGPA) is calculated by multiplying the grade point equivalent of the letter coefficient by the credit units for each course and then dividing the total sum by the total credit units taken in the programme.

Grading System Over 100			Literal Grading System	Grading System Over 4.00
For Non-Technical Courses	For Technical Courses	For Foreign Language Courses		
90-100	80-100	95-100	AA (A)	4.0
85-89	75-79	90-94	BA (B+)	3.5
80-84	70-74	85-89	BB (B)	3.0
75-79	65-69	80-84	CB (C+)	2.5
70-74	60-64	75-79	CC (C)	2.0
65-69	55-59	70-74	DC (D+)	1.5
60-64	50-54	65-69	DD (D)	1.0
0-59	0-49	0-64	F	0

4.5. Overall classification of the qualification

Genel Not Ortalaması : PEKİYİ (3.74 / 4.00)

Cumulative Grade Point Average : OUTSTANDING (3.74 / 4.00)

Graduates whose CGPA are between 2,00-2,49 are listed as "ORTA" (satisfactory), between 2,50-3,49 "İYİ" (excellent), between 3,50-4,00 "PEKİYİ" (outstanding).

5. INFORMATION ON THE FUNCTION OF THE QUALIFICATION

5.1. Access to further study:
May apply to graduate programmes

5.2. Professional status:
Design of hardware systems and applications. Design of information systems. Design of web services. Development of computer programs. Management of computer systems.

6. ADDITIONAL INFORMATION

6.1. Additional information
İstanbul Teknik Üniversitesi, Öğrenci İşleri Daire Başkanlığı 34469 Maslak İstanbul TURKEY

6.2. Further information sources:
University web site: http://www.itu.edu.tr
Student Information System web site: http://www.sis.itu.edu.tr
The Council of Higher Education web site: www.yok.gov.tr
The Turkish ENIC-NARIC web site: www.enic-naric.net/members.asp?country=Turkey

7. CERTIFICATION OF THE SUPPLEMENT

Registrar

Yıldız BÜYÜKÇOLAK

10th of November,2016

8. INFORMATION ON THE NATIONAL HIGHER EDUCATION SYSTEM

Structure and Degree System

The basic structure of the Turkish National Education System consists of stages of noncompulsory pre-school education; compulsory primary (elementary and middle school) and secondary (high school) education; and higher education. Primary education begins at the age of 5.5 (66 months), lasts eight years and comprises elementary and middle school education, four years each. Secondary education is also four years and divided into two categories as "General High School Education" and "Vocational and Technical High School Education". The entry into these categories is through composite scores obtained from a centralized exam for secondary schools.

Higher education system in Turkey is managed by the Council of Higher Education (CoHE, Yükseköğretim Kurulu-YÖK) which is an autonomous public body responsible for the planning, coordination, governance and supervision of higher education within the provisions set forth in the Constitution of the Turkish Republic and the Higher Education Law. Both state and non-profit foundation universities are founded by law and subjected to the Higher Education Law and to the regulations enacted in accordance with it.

Higher education in Turkey comprises all post secondary higher education programmes, consisting of short, first, second, and third cycle degrees in terms of the terminology of the Bologna Process. The structure of Turkish higher education degrees is based on a two-tier system, except for dentistry, pharmacy, medicine and veterinary medicine programmes which have a one-tier system. The duration of these one-tier programmes is five years (300 ECTS) except for medicine which lasts six years (360 ECTS). The qualifications in these one-tier programmes are equivalent to the first cycle (bachelor's) plus second cycle (master's) degree. Undergraduate level of study consists of short cycle (associate's)-(önlisans derecesi) and first cycle (bachelor's)-(lisans derecesi) degrees which are awarded after successful completion of full-time two-year (120 ECTS) and four-year (240 ECTS) study programmes, respectively.

Graduate level of study consists of second cycle (master's)-(yüksek lisans derecesi) and third cycle (doctorate)-(doktora derecesi) degree programmes. Second cycle is divided into two sub-types named as master without thesis and master with thesis. Master programmes without thesis require 60 to 90 ECTS credits and consist of courses and a semester project. 60 ECTS non-thesis master programmes are exceptional, and exist in a few disciplines. The master programmes with a thesis require 90 to 120 ECTS credits, which consists of courses, a seminar, and a thesis. Third cycle (doctorate) degree programmes are completed having earned a minimum of 180 ECTS credits, which consists of completion of courses, passing a proficiency examination and a doctoral thesis. Specialization in medicine, accepted as equivalent to third cycle programmes are carried out within the faculties of medicine, university hospitals and the training hospitals operated by the Ministry of Health.

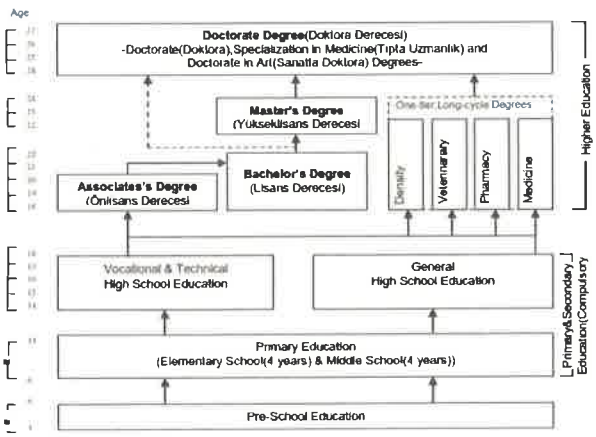
Universities consist of graduate schools (Institutes) offering second cycle (master's) and third cycle (doctorate) degree programmes, faculties offering first cycle (bachelor's degree) programmes, four-year higher schools offering first cycle (bachelor's) degree programmes with a vocational emphasis and two-year vocational schools offering short cycle (associate's) degree programmes of a strictly vocational nature.

Since 2003, first cycle degree holders may apply directly to third cycle (doctorate) programmes if their performance at the first cycle degree level is exceptionally high and their national central Graduate Education Entrance Examination (ALES) score is also high and their application is approved. For these students, theoretical part of the programmes requires additional courses of 60 ECTS credits.

Admission of national students to short and first cycle degree programmes is centralized and based on a nationwide one/two-stage examination(s) conducted by an autonomous public body (Assessment, Selection and Placement Centre-ÖSYM). Candidates gain access to institutions of higher education based on their composite scores consisting of the scores on the selection examination and their high school grade point averages. Admission to graduate programmes is directly conducted by the higher education institutions (HEIs) within the frameworks of the publicly available national and institutional regulations. Admission of foreign students to programmes at all levels of higher education can be done by direct applications of candidates to HEIs based on publicly available national and institutional regulations.

The Turkish National Qualifications Framework for Higher Education (TYYÇ): The National Qualifications Framework for Higher Education in Turkey (TYYÇ) developed with reference to the QF for European Higher Education Area and the EQF for lifelong learning was adopted by the CoHE in 2010. The framework has been developed as a part of a single national qualifications framework, which would eventually consists of 8 level national framework covering all levels of educations on completion of the ongoing work at the national level, in which the higher education levels lie on levels between 5 to 8. The levels of the TYYÇ with reference to the European overarching qualifications frameworks as well as that to ECTS credits and student workload are shown below.

TYYC LEVELS,QUALIFICATIONS TYPES AND ECTS CREDITS

Higher Education Levels/Cycles			AWARDS/ DEGREES	LENGTH (Year)	TOTAL ECTS CREDITS (Year×60 ECTS)	TOTAL STUDENT WORKLOAD(h) (1 ECTS=25-30h)
QF-EHEA	EQF-LLL	TYYÇ LEVELS				
3	8	8	Doctorate Specialization in Medicine Doctorate in Art	3 (min.)	180 (min.)	4.500-5.400
2	7	7	Master's Degree	1-2	60-120	1.500-3.600
1	6	6	Bachelor's Degree	4	240	6.000-7.200
Short Cycle	5	5	Associate's Degree	2	120	3.000-3.600

GENERAL STRUCTURE OF THE TURKISH EDUCATION SYSTEM